AGENTIC AI, GIS & DIGITAL TWIN

Aqua Genesis

Autonomous flood mitigation via real-time hydrologic digital twins.

github.com/tmthangaraj1980-AIML/Aqua_Genesis_project

THE PROBLEM

Reactive Hydrology

Delayed Response: Manual gate control during flash floods is too slow.

Static Mapping: Historical flood maps fail in dynamic storm events.

Complex Physics: Non-linear water flow exceeds human predictive capacity.



System Objectives

GIS Integration

Injest high-resolution DEM and live sensor streams.

Digital Twin

Real-time fluid dynamic simulation of urban topology.

Agentic Control

Autonomous RL agents recommending mitigation acts.

Technology Stack



Python



RLlib



QGIS / GDAL



Hydro-Engine

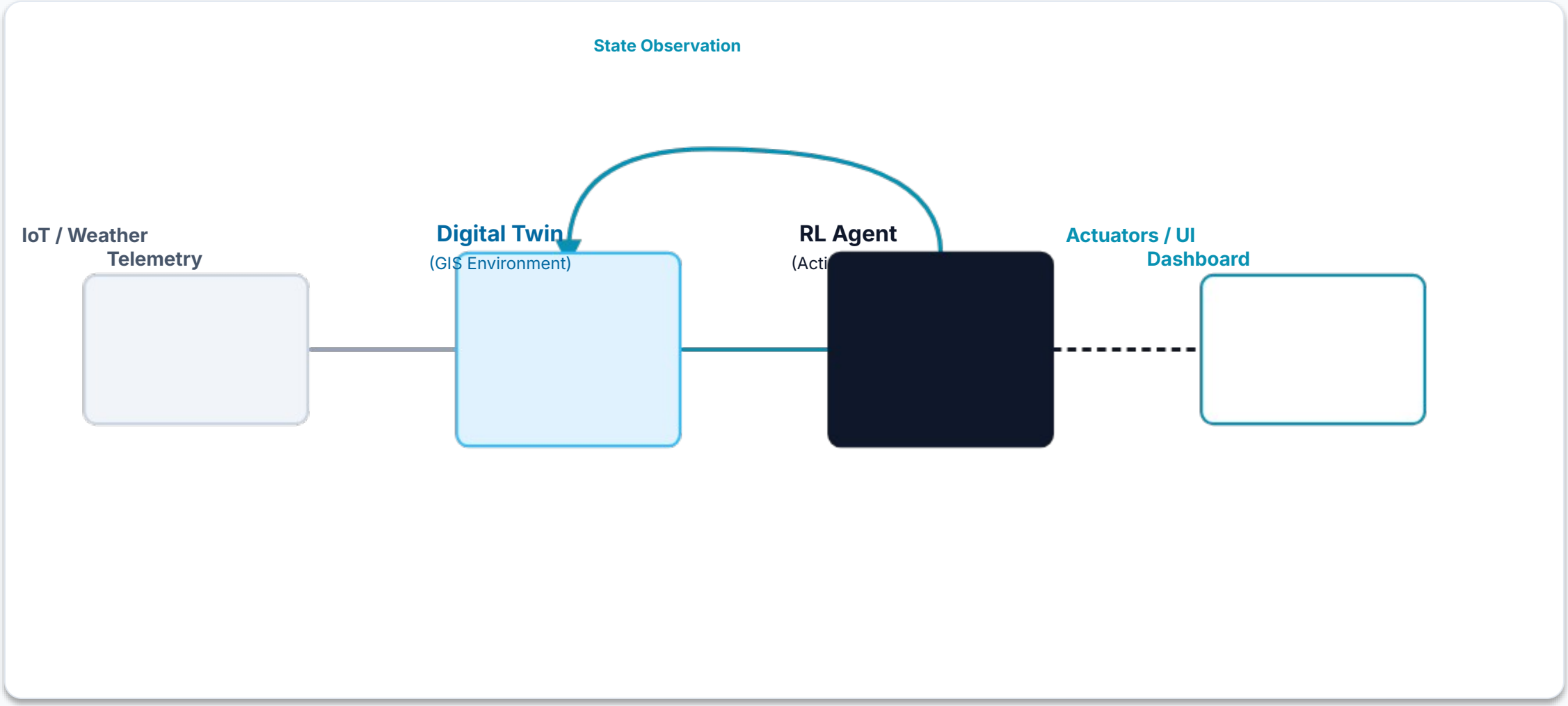


IoT MQTT



Docker

Twin & Agent Architecture



Autonomous Loop



1. Sense

Ingest IoT river gauges.



2. Synchronize

Update 3D Twin state.



3. Decide

Agent selects gate actions.



4. Actuate

Dispatch hardware command.

Data Foundations

Spatial & Temporal Data

- High-Res DEM (Digital Elevation).
- Live rainfall intensity APIs.
- Sewer capacity network nodes.

Environment Preprocessing

- Spatial interpolation of missing telemetry.
- Resolution matching across raster layers.
- State normalization for RL convergence.

Agentic AI Design

State Space

Water depth matrix, gate positions, and impending rainfall vectors.

Action Space

Continuous/discrete variables governing reservoir valve flow rates.

Reward Function

Penalty for flooded urban cells; penalty for excessive gate toggling.



Digital Twin Visualization

Real-time 3D overlay mapping the Agent's simulated water routing decisions against city topology.

Simulation Active

Challenges & Realities

Sim2Real Gap

Friction Coefficients Urban debris dynamically alters real-world water flow rates.

Action Latency Physical dam gates take minutes to actuate, unlike instant simulation.

Compute Bottlenecks

Navier-Stokes Complexity Full hydrodynamic solving is too slow for RL training loops.

Mitigation Relying on simplified cellular automata for rapid state approximation.

Technical Q&A

1. What defines Agentic AI here vs Standard ML?

Standard ML predicts floods; Agentic AI takes actions to prevent them.

2. Purpose of the Digital Twin?

Provides a safe, accelerated, physics-based environment for RL training.

3. Why Reinforcement Learning over hardcoded rules?

Handles unprecedented edge-cases and non-linear interactions better.

4. How do you handle Sim2Real gap?

Domain randomization in training (varying friction and sensor noise).

5. Addressing safety constraints?

Implementing constrained RL paradigms where dangerous actions trigger massive penalties.

Proactive Cyber-Physical Resilience.

Aqua Genesis merges physical infrastructure with agentic intelligence to defend tomorrow's cities.



/Aqua_Genesis_project



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Image Sources



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